

name: _____

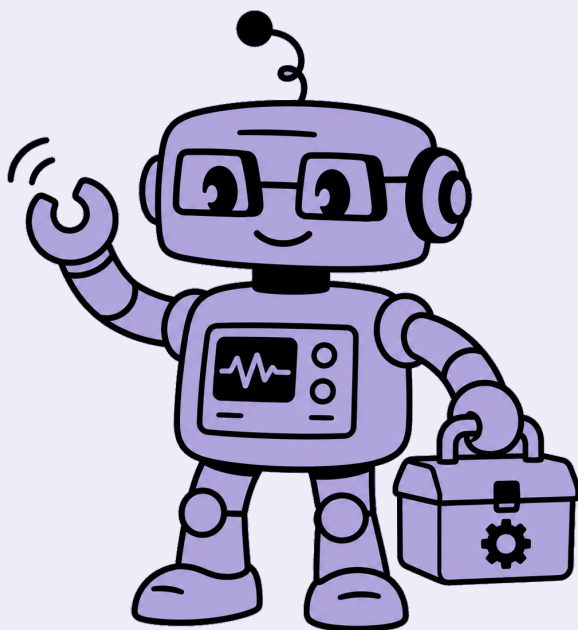
openhouse

robotics • level 2

✦ electronics ✦

your experience & progress book

ages 8-12



the four skills

the skills you grow

in every machine you grow the same four skills. look for the little icon at the top of each page – it tells you which skill you're using.



building & making

you build a real machine from blocks and circuit cards, step by step.



observing & understanding

you watch what your circuit does – then work out *why* it works.



problem solving

you think up and design your own ways to make it better.



presenting

you show your machine and explain how it works.

table of contents

	machine	the big idea	page
1	railway barrier	open & closed circuits	p.5
2	wind turbine	generators, voltage & polarity	p.13
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4	cleaning bot	parallel vs series circuits	p.29
5	sensor-controlled crane	input, signal & output	p.37

each machine takes you through all four skills – build it, watch it, understand it, improve it, and present it.

before you begin

staying safe with electricity

electricity is a helper, but we treat it with care. four rules for every engineer:



the battery blocks in your kit are **safe** to touch and build with – never use wall plugs or house electricity.



never put wires or blocks in your mouth, or near water.



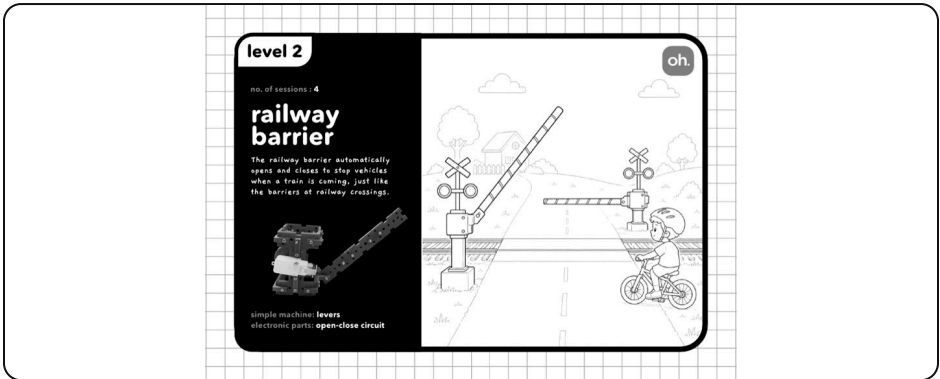
if a wire feels warm, switch it off and tell your teacher.



always build with your teacher nearby – engineers work safely.



meet the railway barrier



your railway barrier – it lifts and lowers when you flip the switch.

Every morning, Aarav's school bus stops at the railway crossing. A striped barrier comes down, a bell rings, and everyone waits as the train rushes past. Then the barrier lifts and the bus rolls on. how does it know when to go up and down? today, think like an engineer.

words to know

circuit

switch

current



did you know?

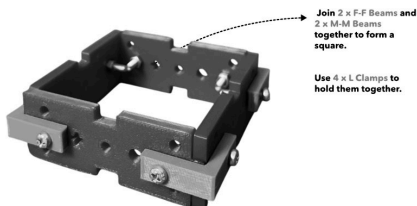
the flow of electricity going round the loop is called **current**.



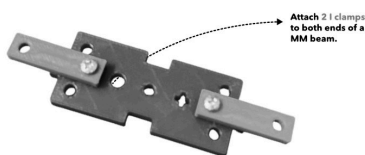
building & making

build your railway barrier – follow the steps.

step 1 : assemble the chassis

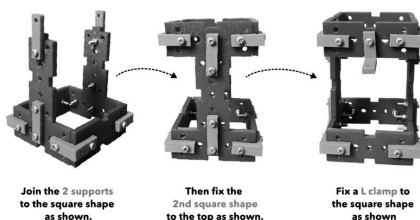


step 2: assemble the supports



Repeat the step to make another one.

step 3: join the supports to the chassis



☐ tick when you've built the frame

next: add the barrier arm and the switch, so it lifts and lowers when you flip the switch.



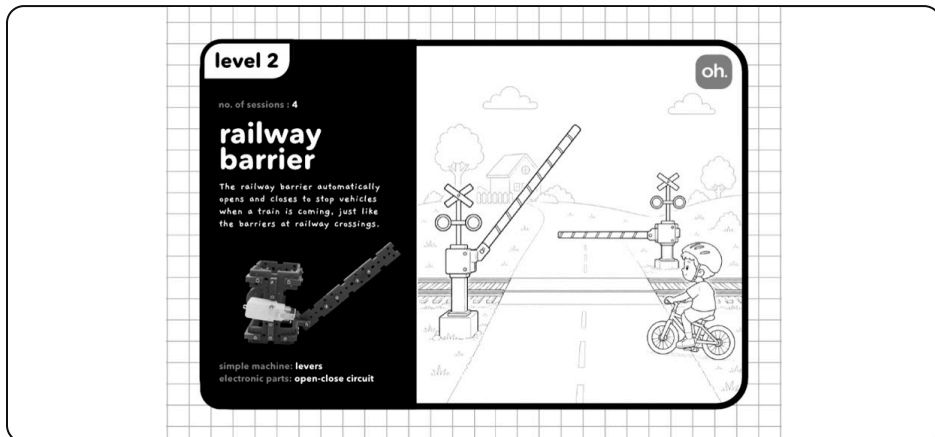
building & making



name the parts



on the picture, circle each part and write its name next to it.



label these four parts

the motor

the battery

the switch

the barrier arm



bonus: draw an arrow to show how it moves.

observing & understanding

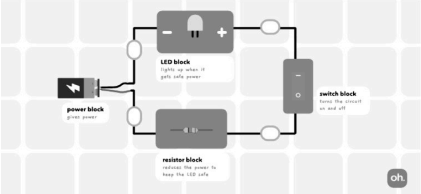
build this circuit – then test it like a scientist.

●

circuit 2 : switch block

oh, easy

Task: Connect all blocks in a circuit and place the switch such that the LED turns on when the switch is ON.



power block
gives power

LED block
lights up when it
gets safe power

resistor block
resists the power to
keep the LED safe

switch block
closes the circuit
on and off

oh

your test

test it 5 times. each time you close the switch, does the light come on? record it.

try	switch closed?	light on?
1		
2		
3		
4		
5		

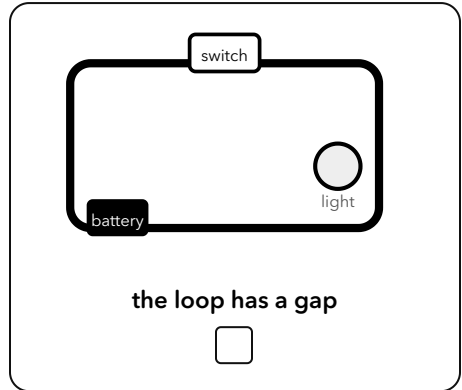
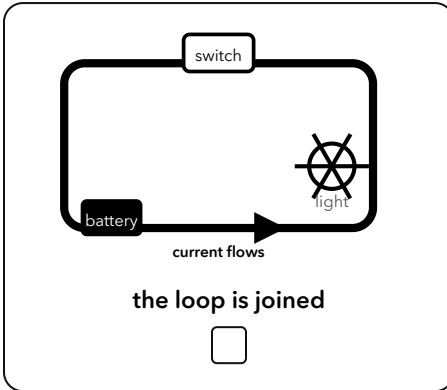
 what does your table show? write one sentence.





observing & understanding

the big idea – why does it work?



a switch is simply a **gap in the loop that you control**. loop joined = a **closed circuit** → current flows. a gap = an **open circuit** → no current. current only ever flows in a complete, closed loop.

in your own words

explain the big idea to a friend – write it in one or two sentences.





problem solving



design challenge

real engineers try more than one idea, then compare.

1 • design a barrier where **ONE** switch controls **TWO** lights

draw a circuit that does this.

2 • now design a second way to do the same thing

draw it below.

3 • compare your two designs

which one uses less wire? circle it and say why.



draw and label your designs here



problem solving



engineer challenge – draw the circuit diagram

draw a **circuit diagram** (using the symbols above) for a barrier light controlled by a switch.

use these circuit symbols



battery



switch



lamp



motor



wire



draw your circuit as a proper diagram – a **complete loop** using the symbols above.

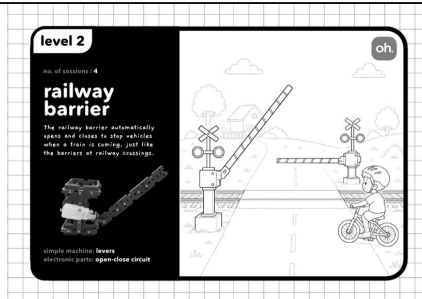


draw your circuit diagram here



presenting

can you be the teacher?



show a partner your railway barrier. explain what it does, which part gives it power, and which part makes it work – and use the right words.

★ **challenge:** show them how to raise and lower it.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it keeps cars and people safe at the railway crossing.)

family member: _____

signature: _____



meet the wind turbine



your wind turbine – it spins in the wind and lights up an led.

On a windy hill, Meera watches a giant white windmill turn, slow and steady. Her teacher says it isn't grinding grain – it's making electricity for the whole village. Meera wonders: how can spinning blades light up a bulb? today, you'll find out.

words to know

energy

led

generator



did you know?

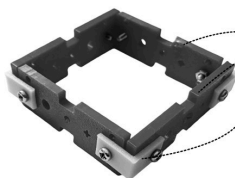
a spinning motor can **make** electricity – then it's called a **generator**.



building & making

build your wind turbine – follow the steps.

step 1 : assemble the base

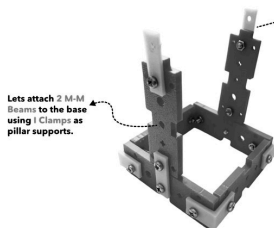


Let us start with building the base of the Wind turbine.

Join
2 x F-F Beam and
2 x M-M Beams
together to form a square.

Use 4 x L-Clamps to hold
them together and
tighten it with screws
and nuts.

step 2 : assemble the pillar supports



Lets attach 2 M-M
Beams to the base
using 1 Clamps as
pillar supports.

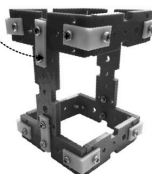
Fix the other set of
i-clamps to the top
of the pillars.

step 3: assemble the 2nd base support

repeat step 1 and
make the second base



now, fix the 2nd
base structure to
the top of the pillar
onto the i-clamps



tick when you've built the frame

🔧 next: connect the motor to the led block, so it spins in the wind and lights up an led.



building & making



name the parts

👉 on the picture, circle each part and write its name next to it.



label these four parts

the blades

the motor

the led

the wires



bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit – then test it like a scientist.

circuit 1 : resistor block & led block

Task: Connect all blocks as per the circuit below such that the LED lights up when it gets safe power.

oh, easy

oh



your test

spin the blades at three speeds. rank the led brightness from 1 (dimnest) to 3 (brightest).

speed	brightness rank (1-3)



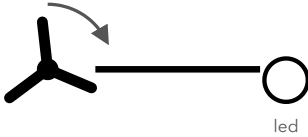
what does your table show? write one sentence.



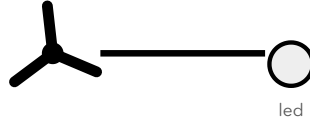


observing & understanding

the big idea – why does it work?



spinning fast



barely spinning

the spinning motor is a **generator**: motion in → electricity out. a faster spin makes a higher **voltage**, so more current flows and the led is brighter. an led has **polarity** – it only lets current through one way, so it must face the right way to light.



in your own words

explain the big idea to a friend – write it in one or two sentences.





problem solving



design challenge

real engineers try more than one idea, then compare.

1 · design a turbine that lights the led in the LOWEST wind

list two changes you could make.

2 · predict which change helps more, then test it

write your prediction, then the result.

3 · keep only the change that worked

draw your best design.



draw and label your designs here



problem solving



engineer challenge – draw the circuit diagram

draw a **circuit diagram**: the generator (motor) connected to the led, showing which way the led must face.

use these circuit symbols



battery



switch



lamp



motor



wire



draw your circuit as a proper diagram – a **complete loop** using the symbols above.



draw your circuit diagram here



presenting

can you be the teacher?



show a partner your wind turbine. explain what it does, which part gives it power, and which part makes it work – and use the right words.

★ **challenge:** show them how to spin it to light the led.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it turns wind into electricity to light a lamp.)

family member: _____

signature: _____



meet the soccer bot

oh.



your soccer bot – it drives forward and back when you flip the switch.

It's the big match! Kabir grips his controller as his little robot races across the floor, nudges the ball, and – GOAL! But to defend, he has to make it reverse, fast. how can one robot go both forwards and backwards? that's your challenge.

words to know

motor

direction

reverse



did you know?

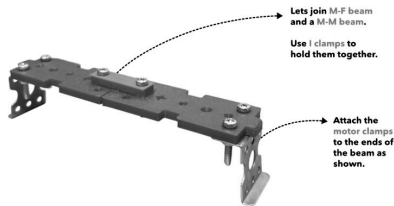
swap the two wires on a motor and it spins the **other** way – that's **reversing**.



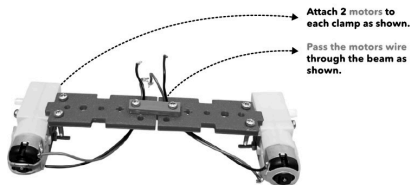
building & making

build your soccer bot – follow the steps.

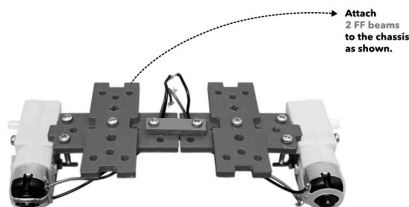
step 1 : assemble the chassis



step 2: attach the motors



step 3: complete the chassis assembly



tick when you've built the frame

🔧 next: wire the motors through the dpdt switch, so it drives forward and back when you flip the switch.

soccer bot



for the child



building & making



name the parts

👉 on the picture, circle each part and write its name next to it.

oh.



label these four parts

the motor

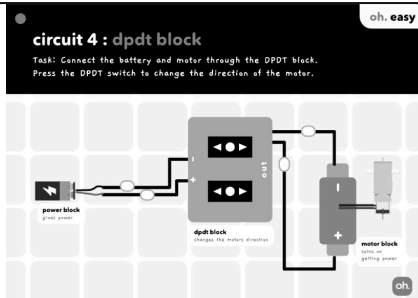
the wheels

the battery

the switch



bonus: draw an arrow to show how it moves.



flip	switch position	wheel direction
1		
2		
3		
4		



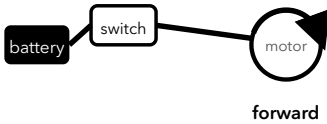
what does your table show? write one sentence.





observing & understanding

the big idea – why does it work?



forward

switch one way



backward

switch the other way

a motor's direction depends on which way the **current** flows through it. swapping the two connections reverses the current, so the motor reverses. a **dpdt switch** swaps both connections at once for you.

in your own words

explain the big idea to a friend – write it in one or two sentences.





problem solving



design challenge

real engineers try more than one idea, then compare.

1 • design controls so the bot turns LEFT and RIGHT

how many motors and switches do you need? draw it.

2 • work out what each switch position should do

fill a small table of your own.

3 • pick the simplest design that still turns both ways

circle it.



draw and label your designs here



problem solving



engineer challenge – draw the circuit diagram

draw a **circuit diagram**: battery → dpdt switch → motor, and note what each switch position does.

use these circuit symbols



battery



switch



lamp



motor



wire



draw your circuit as a proper diagram – a **complete loop** using the symbols above.



draw your circuit diagram here



presenting

can you be the teacher?

oh.



show a partner your soccer bot. explain what it does, which part gives it power, and which part makes it work – and use the right words.

★ **challenge:** show them how to drive it forward and back.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (you drive it to chase the ball and score goals.)

family member: _____

signature: _____



meet the cleaning bot



your cleaning bot – it drives along and sweeps at the same time.

Diya's room is a mess again! She wishes for a little robot that could drive around AND sweep at the same time. Her uncle smiles: one battery can do more than one job at once. how? today, you'll build it and see.

words to know

power

parallel

current



did you know?

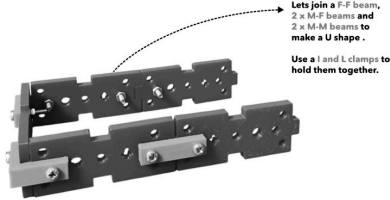
when one battery runs two things at once, we say they are wired in **parallel**.



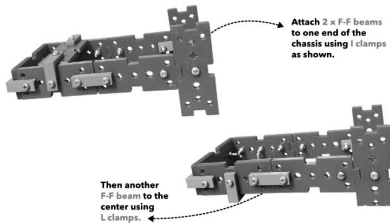
building & making

build your cleaning bot – follow the steps.

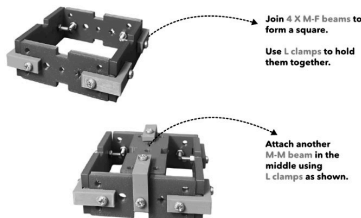
step 1 : assemble the chassis



step 2: attach the supports



step 3: assemble the front chassis



tick when you've built the frame

🔧 next: wire both motors to the one battery, so it drives along and sweeps at the same time.

cleaning bot



for the child



building & making



name the parts

👉 on the picture, circle each part and write its name next to it.



label these four parts

the wheels' motor

the brush's motor

the battery

the brush

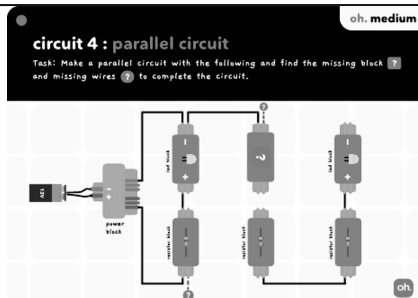


bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit – then test it like a scientist.



 your test

run only the wheels, then only the brush, then both. does the brush slow down when both run? note what you see.

what's running	brush speed (fast/ok/slow)



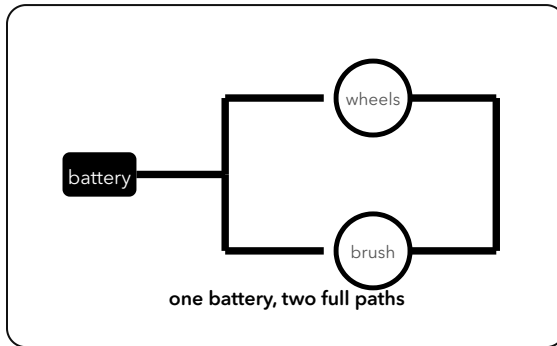
what does your table show? write one sentence.





observing & understanding

the big idea – why does it work?



wiring the two motors in **parallel** gives each its own full path to the battery, so both get full power. wired in **series** (one after the other) they would share the power and both run weaker.

in your own words

explain the big idea to a friend – write it in one or two sentences.





problem solving



design challenge

real engineers try more than one idea, then compare.

1 • design a bot that runs wheels + brush + a light, all from one battery

draw it wired in parallel.

2 • work out how many paths leave the battery

label them on your drawing.

3 • compare parallel vs series for your bot

which keeps everything strong? say why.



draw and label your designs here



problem solving



engineer challenge – draw the circuit diagram

draw a **circuit diagram**: one battery with two parallel branches – a wheels motor and a brush motor.

use these circuit symbols



battery



switch



lamp



motor



wire



draw your circuit as a proper diagram – a **complete loop** using the symbols above.



draw your circuit diagram here



presenting

can you be the teacher?



show a partner your cleaning bot. explain what it does, which part gives it power, and which part makes it work – and use the right words.

★ **challenge:** show them how to drive and sweep at once.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it sweeps the floor while it drives around.)

family member: _____

signature: _____



meet the sensor-controlled crane



your sensor-controlled crane – it lifts the load when the sensor is triggered.

At the port, huge cranes lift heavy boxes off the ships. Rehan notices something clever – the crane seems to KNOW when a box is under it, and only then does it lift. how can a machine sense and then act all by itself? let's build one.

words to know

sensor

input

output



did you know?

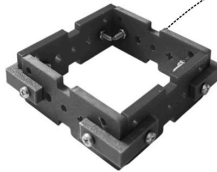
a machine part that **senses** the world (like light or an object) is called a **sensor**.



building & making

build your sensor-controlled crane – follow the steps.

step 1 : assemble the base

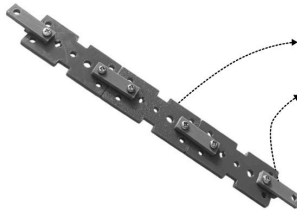


Let us start by building the base of the Crane.

Join 2 x F-F Beam and 2 x M-M Beam together to form a square.

Use 4 x U Clamps to hold them together.

step 2: assemble the supports



Join 3 x M-F Beam together using 1 clamp as shown.

Attach two extra 1 clamps to the ends of the pillar as shown

 Repeat this step to make another set.

step 3: attach the supports to the base



Attach the 2 supports to the base as shown.



tick when you've built the frame

 next: wire the sensor to the motor, so it lifts the load when the sensor is triggered.



building & making



name the parts



on the picture, circle each part and write its name next to it.

oh.



label these four parts

the sensor

the motor

the hook

the battery



bonus: draw an arrow to show how it moves.

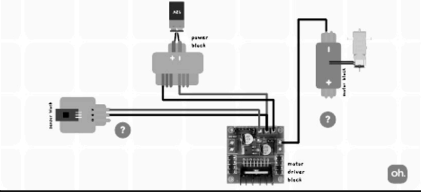
 observing & understanding

build this circuit – then test it like a scientist.

oh, difficult

circuit 1: ir sensor + motor block

Task: Use a motor driver to control a motor block. Control the direction of the motor by changing Out PIN of the sensor block. Can you complete the missing wire ⑦ in the circuit and change the direction of the motor using the sensor?



 **your test**

move an object toward the sensor. measure the distance where the motor starts. try 3 times.

try	distance it triggers (cm)
1	
2	
3	

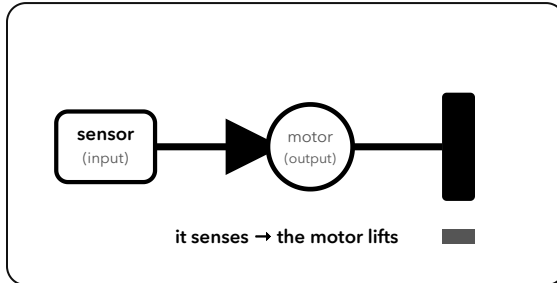
 what does your table show? write one sentence.





observing & understanding

the big idea – why does it work?



the sensor is an **input** – it detects something and sends a **signal**. the motor is the **output** – it acts on that signal. every smart machine works this way: **input** → **decide** → **output**.



in your own words

explain the big idea to a friend – write it in one or two sentences.





problem solving



design challenge

real engineers try more than one idea, then compare.

1 · design a crane that lifts when a box is near AND stops at the top
what inputs and outputs do you need?

2 · draw the input → output flow for your crane
label each part.

3 · add one more sensor to make it safer
say what it would sense.

 draw and label your designs here



problem solving



engineer challenge – draw the circuit diagram

draw a **block diagram**: sensor (input) → controller → motor (output).

use these circuit symbols



battery



switch



lamp



motor



wire



draw your circuit as a proper diagram – a **complete loop** using the symbols above.



draw your circuit diagram here



presenting

can you be the teacher?

oh.



show a partner your sensor-controlled crane. explain what it does, which part gives it power, and which part makes it work – and use the right words.

★ **challenge:** show them how to trigger the sensor to lift the load.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it senses a load and lifts it, all by itself.)

family member: _____

signature: _____